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Abstract

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1. Introduction

This internship is focused on understanding numerical methods for hyperbolic systems of equations, especially for the euler equations. The final objective is to implement a suite of test that trigger a numerical instability at supersonic speeds.

1.1. The Euler equations

Describing the dynamics of fluids by each particle composing the medium is too difficult. Instead we prefer using macroscopic quantities that result from taking the mean of microscopic values over many particles.

For the dynamics of compressible fluids these macroscopic quantities are :

- the **density** $\rho(\mathbf{x}, t)$ in $\text{kg} \cdot \text{m}^{-3}$ resulting from taking the mean mass per volume.
- the **velocity vector** $\mathbf{v}(\mathbf{x}, t) = (u \ v \ w)^T$ in $\text{m} \cdot \text{s}^{-1}$ resulting from taking the mean velocity of each particle per volume.
- the **pressure** $p(\mathbf{x}, t)$ in $\text{Pa} = \text{N} \cdot \text{m}^{-2}$ resulting from the mean magnitude of force per area.

These are called the *primitive* variables, but to compute them we use the *conserved variables* :

- the **density** $\rho(\mathbf{x}, t)$
- the **momentum vector** $\rho \mathbf{v}(\mathbf{x}, t)$
- the **total energy per unit volume** $E(\mathbf{x}, t)$

Using the fundamental laws of conservation of mass, Newton second law and the conservation of Energy we get the Euler equations governing inviscid compressible fluids on the *conserved variables* :

$$\begin{cases} \rho_t + \nabla \cdot (\rho \mathbf{v}) = 0 \\ (\rho \mathbf{v})_t + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + p \mathbf{I}) = 0 \\ E_t + \nabla \cdot [(E + p) \mathbf{v}] = 0 \end{cases} \quad (1.1)$$

where $E = \rho(\frac{1}{2} \mathbf{v} \cdot \mathbf{v} + e)$, with e the **specific internal energy** that is determined using thermodynamical laws (EOS). We will study ideal gases, for which we have $e(\rho, p) = \frac{p}{(\gamma-1)\rho}$.

The system can be rewritten in conservative form :

$$\mathbf{u}_t + \mathbf{F}(\mathbf{u})_x + \mathbf{G}(\mathbf{u})_y + \mathbf{H}(\mathbf{u})_z = 0 \quad (1.2)$$

$$\text{with } \mathbf{u} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ E \end{pmatrix}, \quad \mathbf{F}(\mathbf{u}) = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uw \\ (E+p)u \end{pmatrix}, \quad \mathbf{G}(\mathbf{u}) = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho vw \\ (E+p)v \end{pmatrix} \text{ and } \mathbf{H}(\mathbf{u}) = \begin{pmatrix} \rho w \\ \rho uw \\ \rho vw \\ \rho w^2 + p \\ (E+p)w \end{pmatrix}$$

$\mathbf{u}$ is the column vector of conserved variables and $\mathbf{F}, \mathbf{G}, \mathbf{H}$ are the *flux vectors* in the x, y, z directions respectively.

This is a system of differential equations and thus assumes smooth solutions (partial derivatives exist). In order to handle discontinuous solutions we will need to rewrite it in integral form see Section 2.1.

1.2. Mathematical analysis of the Euler equations

1.3. Submodels

1.3.1. Barotropic Euler equations

By assuming that the density depends only on pressure, $\rho = \rho(p)$, the system simplifies a lot. These are a class of subsystems called the Barotropic Euler equations (from the greek “Baro-” : pressure, “-tropic” : depends on). One particular barotropic system arises when we suppose the entropy stays constant, this is called the *isentropic* Euler equations.

$$\begin{aligned} dS &= 0 \\ \dots \\ p(\rho) &= \kappa \rho^\gamma \end{aligned} \tag{1.3}$$

Another Barotropic system comes from the SWE: explain free surface gravity flows + swe simplifications

$$U = \begin{pmatrix} \varphi \\ \varphi u \end{pmatrix} \text{ and } F(U) = \begin{pmatrix} \varphi u \\ \varphi u^2 + \frac{1}{2} \varphi^2 \end{pmatrix}$$

we recognize a isentropic euler equation with $\kappa = \frac{1}{2}$ and $\gamma = 2$.

1.3.1.1. Analysis of the Isentropic Euler equations

$$U = \begin{pmatrix} \rho \\ \rho u \end{pmatrix} \text{ and } F(U) = \begin{pmatrix} \rho u \\ \rho u^2 + \frac{1}{2} \rho^2 \end{pmatrix}$$

The speed of sound is given by

$$a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s} = \sqrt{\frac{dp}{d\rho}} = \sqrt{\kappa \gamma \rho^{\gamma-1}} \tag{1.4}$$

We now study the eigenvalues and eigenvectors of the jacobian of the flux.

$$\nabla_U F(U) = \begin{pmatrix} 0 & 1 \\ a^2 - u^2 & 2u \end{pmatrix} \tag{1.5}$$

Its eigenvalues are the solutions to the characteristic polynomial given by

$$\lambda^2 - 2u\lambda + u^2 - a^2 = 0 \tag{1.6}$$

As $\Delta = 4a^2 > 0$, the Jacobian has two real roots

$$\lambda_{1,2} = u \pm a \quad (1.7)$$

The corresponding right eigenvectors are

$$K_{1,2} = \begin{pmatrix} 1 \\ u \pm a \end{pmatrix} \quad (1.8)$$

Let us now characterize these fields :

$$\begin{aligned} \text{Let } U \in \mathbb{R}^2, \nabla \lambda_1(U) \cdot K_1(U) &= \begin{pmatrix} \frac{-u}{\rho} - \frac{(\gamma-1)a}{2\rho} \\ \frac{1}{\rho} \end{pmatrix} \cdot \begin{pmatrix} 1 \\ u - a \end{pmatrix} = -\frac{(\gamma+1)a}{2\rho} \neq 0 \\ \text{and } \nabla \lambda_2(U) \cdot K_2(U) &= \begin{pmatrix} \frac{-u}{\rho} + \frac{(\gamma-1)a}{2\rho} \\ \frac{1}{\rho} \end{pmatrix} \cdot \begin{pmatrix} 1 \\ u + a \end{pmatrix} = \frac{(\gamma+1)a}{2\rho} \neq 0 \end{aligned} \quad (1.9)$$

So both field are genuinely non linear. Now we want to solve the Riemann problem for the isentropic Euler equations.

$$U(x, t = 0) = \begin{cases} U_l & \text{if } x \leq x_m \\ U_r & \text{if } x > x_m \end{cases} \quad (1.10)$$

Applying the Rankine Hugoniot condition across the left wave of speed S_-

$$\begin{cases} \rho_l u_l - \rho_* u_* = S_l(\rho_l - \rho_*) \\ \rho_l u_l^2 + \kappa \rho_l^\gamma - \rho_* u_*^2 - \kappa \rho_*^\gamma = S_l(\rho_l u_l - \rho_* u_*) \end{cases} \quad (1.11)$$

and to the right wave :

$$\begin{cases} \rho_* u_* - \rho_r u_r = (u - a)(\rho_* - \rho_r) \\ \rho_* u_*^2 + \kappa \rho_*^\gamma - \rho_r u_r^2 - \kappa \rho_r^\gamma = (u - a)(\rho_* u_* - \rho_r u_r) \end{cases} \quad (1.12)$$

2. Numerical methods for hyperbolic systems

Following the book on Riemann Solvers by Toro [1] we will implement the Godunov method. It is designed to solve hyperbolic systems of partial differential equations, like the Euler system. We will start by looking into the method as presented in the book, before implementing it in the Julia programming language.

2.1. The finite volumes method

The finite volume method results from considering the integral form of conservation laws:

$$\frac{d}{dt} \int_V u \, dv + \int_\Sigma \mathcal{H} \cdot n \, d\sigma = 0 \quad (2.13)$$

where $\mathbf{u}$ is the conserved quantity vector, V the control volume, Σ the boundary of V , $\mathcal{H}$ the flux tensor and $\mathbf{n}$ the outward pointing normal to Σ . This is then enforced on each cell of a given discretization of the domain.

Consider a conservation law :

$$\mathbf{u}_t + \sum_{i=1}^m F_i(\mathbf{u})_{x_i} = 0 \quad (2.14)$$

the integral form gives :

$$\frac{d}{dt} \int_V \mathbf{u} dv = - \sum_{\Sigma_k \in \Sigma} \int_{\Sigma_k} \sum_{i=1}^m F_i(\mathbf{u}) n_i d\sigma \quad (2.15)$$

Where (Σ_i) is a set of faces (or edges in 2D) such that $\cup \Sigma_i = \Sigma$ and $\Sigma_i \cap \Sigma_j = \emptyset$.

It is left to choose a numerical approximation for the integral, a timestepping method and most importantly a way to compute the fluxes.

2.2. Godunov's method

In 1959, Godunov developed a first-order finite volume method for non-linear hyperbolic conservation laws like the Euler system [2]. Hyperbolicity means that the jacobians $\nabla F_i(\mathbf{u})$ have m real eigenvalues and m linearly independent right eigenvectors. It ensure the conservation law is well posed.

The method considers piecewise constant approximation using the cell averages $\bar{\mathbf{u}} = \frac{1}{|V|} \int_V \mathbf{u} dV$. To compute the fluxes it uses an exact one dimensional Riemann solver, that is an algorithm that can find $\mathbf{u}^*(x, t)$ solution of the Riemann problem:

$$\begin{cases} \mathbf{u}_t + F(\mathbf{u})_x = 0 \\ \mathbf{u}(x, 0) = \begin{cases} \mathbf{u}_L & \text{if } x < 0 \\ \mathbf{u}_R & \text{if } x > 0 \end{cases} \end{cases} \quad (2.16)$$

where the left and right states $\mathbf{u}_L$ and $\mathbf{u}_R$ are the cell averages of the two neighboring cells separated by the face. The flux is then given by $F(\mathbf{u}^*(0, t))$

Conceptually we are letting the fluid flow for some small timestep before taking the average of the cell. As we add and subtract the same amount, the flux, between two cells, this method is conservative.

To determine the timestep we use the criteria that the fastest wave should not travel more than one cell in one timestep. Supposing that no wave acceleration takes place as a consequence of wave interaction, for structured meshes this translates to :

$$\Delta t \leq \frac{\Delta x_i}{\max(|\lambda|)} \quad (2.17)$$

with λ the eigenvalues of the jacobian of the flux in the direction x_i . Then using a coefficient $0 < C_{\text{cfl}} < 1$ we obtain :

$$\Delta t = C_{\text{cfl}} \frac{\Delta x_i}{\max(|\lambda|)} \quad (2.18)$$

Godunov only presented this method for structured meshes, where each direction has its own flux function F_i . Under the condition that the system is *rotationally invariant*, this approach is generalized to unstructured meshes. Using the rotation matrix $T(\mathbf{n})$ that transforms vector components aligning them with the normal coordinate axis and leaves the scalar component unchanged, we get :

$$\sum_i F_i(\mathbf{u}) n_i = T(\mathbf{n})^{-1} F_1(T(\mathbf{n})\mathbf{u}) \quad (2.19)$$

This in turn makes it possible to approximate the face flux as $\int_{\Sigma_k} \sum_i F_i(\mathbf{u}) n_i d\sigma \approx |\Sigma_k| T^{-1} \hat{F}$, where $T^{-1} = T(\mathbf{n})^{-1}$. $\hat{F} = F_1(\mathbf{u}_{\text{rot}}^*)$ is the numerical flux obtained using $\mathbf{u}_{\text{rot}}^*$ the solution of a one dimensional Riemann solver in the rotated coordinates.

For a forward in time approximation, the update would then read :

$$\bar{\mathbf{u}}_i^{n+1} = \bar{\mathbf{u}}_i^n - \frac{\Delta t}{|V_i|} \sum_{\Sigma_k \in \Sigma} |\Sigma_k| T^{-1} \hat{F} \quad (2.20)$$

2.3. Exact Riemann solvers

Now lets see how to solve the Riemann problem (2.16). We are considering hyperbolic conservation laws, so the Jacobian $\nabla F(\mathbf{u}) = \begin{pmatrix} \frac{\partial u_1}{\partial u_1} & \dots & \frac{\partial u_1}{\partial u_n} \\ & \ddots & \\ \frac{\partial u_n}{\partial u_1} & \dots & \frac{\partial u_n}{\partial u_n} \end{pmatrix}$ is diagonalizable with real eigenvalues. Thus there exists an invertible matrix $P(\mathbf{u})$ and a diagonal matrix $D(\mathbf{u})$ such that $\nabla F(\mathbf{u}) = P(\mathbf{u}) D(\mathbf{u}) P(\mathbf{u})^{-1}$.

We call *characteristic variables* the components of the vector

$$\mathbf{v} = P(\mathbf{u})^{-1} \mathbf{u} \quad (2.21)$$

. Using them, the conservation law can be rewritten as :

$$\mathbf{v}_t + D(\mathbf{u}) \mathbf{v}_x = 0 \quad (2.22)$$

which separates the system into n independent equations of the form:

$$\frac{\partial v_i}{\partial t} + \lambda_i(\mathbf{u}) \frac{\partial v_i}{\partial x} = 0 \quad (2.23)$$

Where $\lambda_i(\mathbf{u})$ are the diagonal entries of $D(\mathbf{u})$, i.e. the eigenvalues of $\nabla F(\mathbf{u})$. They are called the *characteristic speeds*.

2.3.1. Riemann solvers for linear systems

When the system is linear, D doesn't depend on $\mathbf{u}$ and we have independent transport equations with an advection speed d_i . Now to find $\mathbf{u}^*$, we have to look at the upwind direction, that is the direction in which information is travelling. If $d_i > 0$ the flow goes

to the right then $u_i^* = u_{L,i}$ on the contrary if $d_i < 0$, $u_i^* = u_{R,i}$. Doing this for all the equations we can construct $\mathbf{u}^*$ and compute the flux as $F(\mathbf{u}^*)$.

Example 2.3.1.1 (The wave equations):

The wave equations is a one-dimensional linear system :

$$\begin{cases} p_t + \frac{1}{\rho}u_x = 0 \\ u_t + \kappa p_x = 0 \end{cases} \quad (2.24)$$

It can be rewritten in conservative form :

$$\mathbf{u}_t + F(\mathbf{u})_x = 0 \quad (2.25)$$

with $\mathbf{u} = \begin{pmatrix} p \\ u \end{pmatrix}$ and $F(\mathbf{u}) = \begin{pmatrix} \frac{1}{\rho}u \\ \kappa p \end{pmatrix}$

Diagonalizing the Jacobian gives :

$$\nabla F(\mathbf{u}) = \begin{pmatrix} 0 & \frac{1}{\rho} \\ \kappa & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ \rho c & -\rho c \end{pmatrix} \begin{pmatrix} c & 0 \\ 0 & -c \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2\rho c} \\ \frac{1}{2} & -\frac{1}{2\rho c} \end{pmatrix} \quad (2.26)$$

with $c = \sqrt{\frac{\kappa}{\rho}}$. We have

$$\mathbf{u}_t + \nabla F(\mathbf{u})\mathbf{u}_x = 0 \quad (2.27)$$

The characteristic variables are :

$$\mathbf{v} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2\rho c} \\ \frac{1}{2} & -\frac{1}{2\rho c} \end{pmatrix} \mathbf{u} = \begin{pmatrix} \frac{p}{2} + \frac{u}{2\rho c} \\ \frac{p}{2} - \frac{u}{2\rho c} \end{pmatrix} \quad (2.28)$$

and we can write :

$$\mathbf{v}_t + \begin{pmatrix} c & 0 \\ 0 & -c \end{pmatrix} \mathbf{v}_x = 0 \quad (2.29)$$

We obtain two independent transport equations for each characteristic variable. One is advected to the right with characteristic speed c and the other is advected to the left with a characteristic speed $-c$. Finally we can use the upwind method and compute the flux.

2.3.2. Riemann Solvers for non-linear systems

A more complicated case arises when the characteristic speeds depend on $\mathbf{u}$. This results in non-linear relationships between the state $\mathbf{u}$ and the flux. Still the approach stays similar to the linear case as we look at the orientation of the flow to determine the upwind direction. But contrary to the linear case, wave distortion can arise from the difference of $d(\mathbf{u})$ between two neighboring points. This can lead to the creation of shock and rarefaction waves.

Shock waves are thin layers where properties change very rapidly. These are generally approximated as mathematical discontinuities. To find the speed S at which this discontinuity is moving, we use the *Rankine-Hugoniot Condition* :

$$\Delta \mathbf{f} = S \Delta \mathbf{u} \quad (2.30)$$

It is obtained by applying the integral form of the conservation law on a control volume encompassing the shock (see [1], p. 70).

On the other hand, rarefaction waves are created through a stretching between two states. To solve it we can't use a discontinuity like in the shock case as this would violate the entropy condition and lead to instabilities. Instead the rarefaction is modelled by two different waves the head and the tail. The solution inside of the rarefaction fan can then be determined using the self similarity condition and the Generalized Riemann invariants.

Example 2.3.2.1 (Burgers' equation):

The Burgers' equation is given by

$$u_t + F(u)_x = 0 \quad (2.31)$$

with $F(u) = \frac{1}{2}u^2$ and $\frac{\partial F(u)}{\partial u} = u$.

Here the propagation speed depends on the value of u and the flow is non-linear.

include image burgers

To solve the Riemann problem (2.16), we have to distinguish between the shock and the rarefaction case.

When $u_L > u_R$ we have a shock. To compute its speed and direction, we use the *Rankine-Hugoniot Condition* (2.30):

$$\frac{1}{2}(u_R^2 - u_L^2) = S(u_R - u_L) \quad (2.32)$$

and we obtain the shock wave speed as the mean between the left and right state:

$$S = \frac{1}{2}(u_R + u_L) \quad (2.33)$$

If $S > 0$ then the shock is moving to the right and if $S < 0$ then the shock is moving to the left. This gives us the solution :

$$u(x, t) = \begin{cases} u_L & \text{if } x - St < 0 \\ u_R & \text{if } x - St > 0 \end{cases} \quad (2.34)$$

Now for the case $u_L < u_R$ we get a rarefaction. The head and tail are given by the flux of the left and right states and we interpolate linearly inbetween them. The solution is given by :

$$u(x, t) = \begin{cases} u_L & \text{if } \frac{x}{t} < u_L \\ \frac{x}{t} & \text{if } u_L < \frac{x}{t} < u_R \\ u_R & \text{if } \frac{x}{t} > u_R \end{cases} \quad (2.35)$$

finally we sample the result at $u^* = u(0, t)$ and compute the flux as $f(u^*)$.

For systems, we have to look at the field of each characteristic(2.21). An eigenvalue $\lambda_i(\mathbf{u}(\mathbf{x}, t))$ characterises the speed of a wave in the physical space $(x - t)$. It is associated with a corresponding right eigenvector $\mathbf{K}_i(\mathbf{u})$, that defines the paths of the wave in phase space.

A λ_i -characteristic field is called *linearly degenerate* when

$$\nabla \lambda_i(\mathbf{u}) \cdot \mathbf{k}_i(\mathbf{u}) = 0, \quad \forall \mathbf{u} \in \mathbb{R}^m \quad (2.36)$$

this means that the evolution of this wave doesn't affect its speed. Discontinuities on such field are called contact waves. On the other hand, we call a λ_i -characteristic field *genuinely non-linear* when

$$\nabla \lambda_i(\mathbf{u}) \cdot \mathbf{k}_i(\mathbf{u}) \neq 0, \quad \forall \mathbf{u} \in \mathbb{R}^m \quad (2.37)$$

in that case, the evolution of the wave affects its speed and a discontinuity on such field creates either shocks or rarefactions.

For more details look at the end of chapter 2 of the book by Toro [1] (pp. 76-85).

2.3.3. The exact Riemann Solver for the Euler equations

Now let's study the one-dimensional euler equation in order to determine the solution to the associated Riemann Problems. The conservative formulation (1.2) can be rewritten in quasi-linear form using the Jacobian of the flux $\nabla_{\mathbf{u}} \mathbf{F}(\mathbf{u})$ as

$$\mathbf{u}_t + \nabla_{\mathbf{u}} \mathbf{F}(\mathbf{u}) \mathbf{u}_x = 0 \quad (2.38)$$

Its eigenvalues are

$$\lambda_1 = u - a, \quad \lambda_2 = u \quad \text{and} \quad \lambda_3 = u + a \quad (2.39)$$

and the associated right eigenvectors are

$$\mathbf{k}_1 = \begin{pmatrix} 1 \\ u - a \\ H - ua \end{pmatrix}, \quad \mathbf{k}_2 = \begin{pmatrix} 1 \\ u \\ \frac{1}{2}u^2 \end{pmatrix} \quad \text{and} \quad \mathbf{k}_3 = \begin{pmatrix} 1 \\ u + a \\ H + ua \end{pmatrix} \quad (2.40)$$

where $H = \frac{E+p}{\rho}$ the total specific enthalpy and $a = \sqrt{\frac{\gamma p}{\rho}}$ the sound speed. For the computations see [1] pp. 87-90.

Computing $\nabla \lambda_{i(\mathbf{u})} \cdot \mathbf{k}_i$, we get that the λ_1 and λ_3 fields are genuinely non-linear and that the λ_2 field is linearly degenerate. As $a > 0$, we have that $\lambda_1 < \lambda_2 < \lambda_3$ so the structure of the solution to the Riemann problem will consist of a contact wave surrounded by two non-linear waves (shocks and/or rarefactions).

Accross the contact wave, the pressure and particle velocity are constant and the density jumps discontinuously. Accross rarefaction waves, ρ, u and p change smoothly. Finally accross a shock wave, ρ, u and p change very rapidly.

The solution to the Riemann problem has 4 regions, on the left side of the left non-linear wave, we have $\mathbf{u}_L$, then on the right side of the right nonlinear wave, we have $\mathbf{u}_R$. The region in between the two non linear waves is called the *star region*, it is split in two by the contact wave. Pressure and particle velocity are constant in the star region but density changes accross the contact wave. We thus have to determine the 4 unknowns ρ_L^*, ρ_R^*, u^* and p^* as well as the left and right non linear waves (shock speed and the solution inside the rarefaction fans).

The first step of the exact riemann solver is to determine u^* and p^* . For this we distinguish the cases of shocks and rarefactions on the left and right sides. For shock waves we start by changing the reference frame to align with the shock speed. This results in the shock speed in the reference frame to be null. We can than apply the Rankine-Hugoniot conditions that are simplified :

$$\mathbf{F}(\hat{\mathbf{u}}_K) = \mathbf{F}(\hat{\mathbf{u}}^*), \quad (2.41)$$

with $\hat{\mathbf{u}}$ the conserved variables in the new frame, where only the particle velocities are adapted and K being either left or right. From this we can deduce the relations

$$\begin{cases} u^* = u_L - f_L(p^*, \mathbf{w}_L) & \text{for the left side case} \\ u^* = u_R + f_R(p^*, \mathbf{w}_R) & \text{for the right side case} \end{cases} \quad (2.42)$$

where

$$f_K(p^*, \mathbf{w}_K) = (p^* - p_K) \sqrt{\frac{2}{((\gamma + 1)\rho_K) \left(p^* + \frac{\gamma-1}{\gamma+1} p_K \right)}}, \quad (2.43)$$

is a function that only depends on p^* and the known state $\mathbf{w}_K$ (left or right).

For the Rarefaction cases, we also obtain the same relations as (2.42) but with a different function f_K . Using the isentropic law and the generalised Riemann invariants, one can obtain :

$$f_K(p^*) = \frac{2a_K}{\gamma - 1} \left[\left(\frac{p^*}{p_K} \right)^{\frac{\gamma-1}{2\gamma}} - 1 \right]. \quad (2.44)$$

We obtain two equation of the form (2.42), one for the left non-linear wave and another for the right. Subtracting them yields

$$f_L(p^*, \mathbf{w}_L) + f_R(p^*, \mathbf{w}_R) + u_R - u_L = 0. \quad (2.45)$$

This means that p^* is the root of the function :

$$f(p^*, \mathbf{w}_L, \mathbf{w}_R) = f_L(p^*, \mathbf{w}_L) + f_R(p^*, \mathbf{w}_R) + u_R - u_L \quad (2.46)$$

That can be found using Newton's method as it is differentiable and has a simple behaviour. We can then determine u^* using (2.42). This leaves to determine the densities while determining the non-linear waves. The type of non-linear wave is determined by the difference between the pressure in the star region and the outside pressure. When the pressure in the star region is greater to the exterior one we have a shock. We get the density of the left/right star region from

$$\rho_K^* = \rho_K \left[\frac{\frac{p^*}{p_K} + \frac{\gamma-1}{\gamma+1}}{\frac{\gamma-1}{\gamma+1} \frac{p^*}{p_K} + 1} \right] \quad (2.47)$$

and the shock speed

$$S_L = u_L - a_L \sqrt{\frac{\gamma+1}{2\gamma} \frac{p^*}{p_L} + \frac{\gamma-1}{2\gamma}} \quad (2.48)$$

or

$$S_R = u_R + a_R \sqrt{\frac{\gamma+1}{2\gamma} \frac{p^*}{p_R} + \frac{\gamma-1}{2\gamma}} \quad (2.49)$$

with K being either left or right.

When the pressure in the star region is smaller than the exterior one, we have a rarefaction wave. We get the density from the isentropic law

$$\rho_K^* = \rho_K \left(\frac{p^*}{p_K} \right)^{\frac{1}{\gamma}} \quad (2.50)$$

and the sound speed is

$$a_K^* = a_K \left(\frac{p^*}{p_K} \right)^{\frac{\gamma-1}{2\gamma}} \quad (2.51)$$

For a left rarefaction wave, the speed of the head and tail are given by

$$\begin{aligned} S_{HL} &= u_L - a_L \text{ for the head} \\ S_{TL} &= u_L^* - a_L^* \text{ for the tail} \end{aligned} \tag{2.52}$$

For a right rarefaction wave, the speed of the head and tail are given by

$$\begin{aligned} S_{HR} &= u_R + a_R \text{ for the head} \\ S_{TR} &= u_R^* + a_R^* \text{ for the tail} \end{aligned} \tag{2.53}$$

The solution inside a rarefaction fan can be found using the Generalised Riemann Invariants and the slope of the characteristic emanating from the origin to the point (x, t) inside the fan.

Finally we can sample the solution at $(0, t)$ and compute the flux for our finite volume solver.

This solver is exact but may not be computationally friendly for big problems as we have to solve a root finding problem. Furthermore it doesn't allow for vacuum (division by zero) which can happen when removing all the density of a cell.

2.3.3.1. Approximate Riemann Solvers for the Euler system

2.3.3.2. Exact Riemann solver for the Barotropic Euler equations

We have $U = \begin{pmatrix} \rho \\ \rho u \end{pmatrix}$ and $F(U) = \begin{pmatrix} \rho u \\ \rho u + \kappa \rho^\gamma \end{pmatrix}$.

2.3.4. Managing Boundary conditions

2.3.5. Computing timestep on unstructured meshes

3. Implementation in Julia

The first part of this internship was dedicated to understand and implement the Godunov method. I chose to use Julia, a high level programming language designed for scientific computing.

3.1. Structure of the code

The dependencies of the Julia project are defined in the `Project.toml` file and they are versionned in the `Manifest.toml` file. This makes it easy to use the same packages on other devices.

3.1.1. Source code

The source code is divided in multiple parts, each defining a specific part of the solver. The custom `mesh` structure holds the data of the mesh (points, cells, faces, boundaries) as well as some precomputed values used in the solver loop. Some helper functions makes it easy to generate regular meshes (1D) and load meshes from a `.msh` file. The solver uses types to distinguish between different **equations**. To define an equation one needs to define the number of variables, a numerical flux and a cfl condition. **Boundary conditions** are also defined in structures and are solved using ghost cells. The solver interface then requires a mesh, an equation type and a dictionary linking physical borders to boundary conditions. The solver in itself uses the finite volume method as in Section 2.1. The data is written in `.vtu` files during the simulation. The data to be written is defined by a method `output_fields` for each type of equation.

3.2. Models

Some models are already implemented and can be directly used. We started with linear models like the linear transport and the wave system. For such equations we proceed as described in Section 2.3.1 and compute the numerical flux from the upwind direction. Then we experienced with the non linear Burgers' equation. To compute the numerical flux we proceeded as in Example 2. In comparison to the linear case, we now had to compute the fastest wave dynamically as the state was influencing it. Finally we implemented the full Euler equations for ideal gases using the Godunov method with multiple Riemann solvers.

3.3. Tests

4. Numerical Instabilities at supersonic speeds

4.1. The Carbuncle instability

4.2. Test cases

4.2.1. Slow moving shock

4.2.2. Kinked mach stem (shock on a ramp)

4.2.3. Odd-even decoupling

4.2.4. Supersonic flow around blunt body (half circle)

4.2.5. Supersonic flow around blunt body (ringleb)

4.2.6. Sedov blast wave problem

4.2.7. Forward facing step

4.2.8. Noh implosion problem

4.2.9. Modified quirk tests : advancing shock wave, reflecting shock wave and steady shock wave

4.2.10. Slight perturbation in slowly moving shocks

4.2.11. SWE : Kelvin-Helmholtz instability + small perturbation

4.2.12. Steady circular hydraulic jump

4.2.13. Elling test

4.2.13.1. Elling test for the full Euler equations

Data from Fleischman et al, A low dissipation method to cure the grid aligned shock instability

4.2.13.2. Elling test for the SWE

Let's look at a shock in the SWE. we have two states:

$$\begin{pmatrix} h_l \\ h_l u_l \end{pmatrix} \text{ and } \begin{pmatrix} h_r \\ h_r u_r \end{pmatrix} \quad (4.54)$$

we set ourselves in the shock reference:

$$\widehat{u}_k = u_k - S \quad (4.55)$$

with $S = \text{Fr}_r \sqrt{gh_r}$ the shock speed The rankine Hugoniot conditions give

$$\begin{cases} h_l \widehat{u}_l = h_r \widehat{u}_r \\ h_l \widehat{u}_l^2 + \frac{1}{2} h_l^2 = h_r \widehat{u}_r^2 + \frac{1}{2} h_r^2 \end{cases} \quad (4.56)$$

isolating $\widehat{u}_r^2$ and using the first relation, we get

$$\widehat{u}_r^2 = \frac{1}{2} \frac{h_l}{h_r} (h_r + h_l) \quad (4.57)$$

using $\widehat{u}_r = u_r - S = (\text{Fr}_r - \text{Fr}_l) \sqrt{h}$ we get a quadratic equation for the ratio $\frac{h_l}{h_r}$. We obtain :

$$\frac{h_l}{h_r} = \frac{1}{2} \left(-1 \pm \sqrt{1 + 8\Delta \text{Fr}^2} \right) \quad (4.58)$$

.

Both h_l and h_r are positive so

$$h_l = \frac{h_r}{2} \left(\sqrt{1 + 8\Delta \text{Fr}^2} - 1 \right) \quad (4.59)$$

Finally using the conservation of mass : $\widehat{u}_l = \widehat{u}_r \frac{h_r}{h_l}$, we get

$$\begin{aligned} u_l &= \text{Fr}_s \sqrt{h_r} + \frac{2(\text{Fr}_r - \text{Fr}_s) \sqrt{h_r}}{\sqrt{1 + 8\Delta \text{Fr}^2} - 1} \\ &= \sqrt{h_r} \left(\text{Fr}_s + \frac{1 + \sqrt{1 + 8\Delta \text{Fr}^2}}{4(\text{Fr}_r - \text{Fr}_s)} \right) \end{aligned} \quad (4.60)$$

Example 4.2.13.2.1 (Elling Test for the Shallow Water Equations): The example from [3] sets $h_l = 1, \text{Fr}_l = 30$ and $S = 0$. Here we are on the opposite case, where the left state is given and we have to determine the right one. First we can compute $u_l = \text{Fr}_l * \sqrt{h_l} = 30$. Then we use our formulas to compute h_r and u_r :

$$\begin{aligned}
h_r &= \frac{1}{2} \left(\sqrt{1 + 8 \times \text{Fr}_l^2} - 1 \right) \approx 41.929 \\
u_r &= \frac{h_l}{h_r} u_l \approx 0.71549
\end{aligned} \tag{4.61}$$

In order to trigger the instability a horizontal filament ($u = 0$) is introduced in the middle of the left part as in the Elling test for the Euler equations.

4.2.14. Laminar boundary layer test

Bibliography

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